

Practical Electronics with KiCad

A learn-by-building course — Chapters 1–3

Project: One LED (screw terminal → resistor → LED)

How this book works

Each chapter pairs a small amount of theory with a real KiCad project that proves it. You read the idea, then you build it, then the math on the page becomes something you can measure with a multimeter. The book grows one chapter at a time as the projects do.

Chapter 1 — Electricity fundamentals

1.1 Atoms and electrons

Everything is made of atoms: a positively charged nucleus surrounded by negatively charged electrons. In most materials the electrons are firmly attached to their atoms. In metals — like the copper traces on your PCB — the outermost electrons are so loosely held that they drift freely from atom to atom. A copper wire is best imagined as a pipe that is already completely full of loose electrons, waiting for something to push them.

Key idea: Conductors (copper, gold, solder) have free electrons. Insulators (the green fiberglass of your PCB, plastic wire coating) hold their electrons tightly, so current cannot flow through them.

1.2 Voltage — the push

Voltage is electrical pressure: a difference in energy between two points that pushes electrons through a conductor. The classic analogy is water. A water tank raised high above the ground has pressure; open a valve and water flows. Raise the tank higher (more voltage) and the water flows harder. Crucially, voltage is always a *difference* between two points. Saying "5 volts" only has meaning as "5 volts higher than the reference point."

1.3 GND — the reference point and the return path

Ground (GND) is simply the point in the circuit that everyone agrees to call 0 V — the floor that all heights are measured from. It is also the return path: electricity only flows in a complete loop, out of the supply's positive terminal, through the components, and back into the supply's negative terminal (GND). Break the loop anywhere — a cut trace, a loose screw terminal — and all flow stops instantly.

Key idea: No loop, no current. Every working circuit is a closed loop from + to – through the components.

1.4 Current — the flow

Current is the amount of electric charge flowing past a point per second, measured in amperes (A). One ampere is roughly 6.24×10^{18} electrons per second. Small circuits use milliamps (mA): 1 mA = 0.001 A. Your LED circuit carries about 13.6 mA.

1.5 Conventional current vs. electron flow

A historical quirk worth knowing once and then not worrying about: electrons are negative, so they physically drift from GND toward +5V. But circuit diagrams and all the math use *conventional current*, which flows from + to –. The convention was set before the electron was discovered, the math works identically either way, and every engineer, book, and simulator uses it. This book does too.

1.6 Resistance — the friction

Resistance, measured in ohms (Ω), is how strongly a material opposes the flow of current — a narrow section of pipe in the water analogy. Resistors are components manufactured to have a precise resistance. Your R1 is 220 Ω . Its job in this circuit is to limit the current so the LED does not destroy itself: without it, the LED would draw far too much current and burn out in well under a second.

1.7 No difference, no flow

Conventional current always flows from higher potential to lower potential, like water flowing downhill. A direct consequence: if both terminals of a circuit sat at the same voltage — say GND at 5 V and the supply also at 5 V — the difference would be 0 V and no current would flow at all. Only voltage *differences* matter; absolute values are pure bookkeeping. A circuit running between 105 V and 100 V behaves identically to one running between 5 V and 0 V: the loop only ever sees the 5 V difference. This is also why a bird perches safely on a high-voltage line — both feet are at the same potential, so 0 V appears across the bird and no current passes through it.

Chapter 2 — Ohm's Law and Kirchhoff's Laws

2.1 Ohm's Law — the one equation that rules everything

$$V = I \times R$$

Voltage (volts) equals current (amps) times resistance (ohms). Rearranged: $I = V \div R$ and $R = V \div I$. Memorize all three forms — you will use them constantly. Intuition: more pressure (V) pushes more flow (I); more friction (R) allows less flow.

2.2 Kirchhoff's Voltage Law — the voltage budget

Around any complete loop, the voltage supplied must exactly equal the sum of the voltages used up by the components. The supply provides a 5 V budget; every component in the loop spends part of it; the total spent is exactly 5 V, arriving at GND with 0 V left. Each component's share is called its *voltage drop*.

2.3 The LED's voltage drop

An LED is not a resistor — it behaves like a toll gate. It takes a roughly fixed voltage, called the *forward voltage* (V_f), and converts the energy of the electrons crossing that gap directly into light. Typical values: red ≈ 1.8 – 2.0 V, yellow/green ≈ 2.0 – 2.2 V, blue/white ≈ 3.0 – 3.4 V. We use 2.0 V for a standard red LED.

2.4 Doing the math for your circuit

Step 1 — voltage budget (Kirchhoff): the supply gives 5 V, the LED takes 2 V, so the resistor must drop the remainder:

$$V_{R1} = 5\text{ V} - 2\text{ V} = 3\text{ V}$$

Step 2 — current (Ohm's Law): the resistor has 3 V across its 220 Ω :

$$I = V \div R = 3\text{ V} \div 220\ \Omega = 0.0136\text{ A} \approx 13.6\text{ mA}$$

Step 3 — sanity check: a standard LED is happy between about 5 and 20 mA. 13.6 mA gives good brightness with a comfortable safety margin. This calculation is exactly why 220 Ω is the classic LED resistor for 5 V supplies. Compare: 100 Ω would give 30 mA (too much — shortened LED life), while 1 k Ω gives 3 mA (safe, but dim).

Key idea: Series components share ONE identical current, but each takes its own voltage drop. Current is conserved around the loop; voltage is spent.

2.5 Power — where the energy goes

$$P = V \times I$$

Power (watts) is energy per second. Your resistor dissipates $P = 3\text{ V} \times 0.0136\text{ A} \approx 0.041\text{ W}$ (41 mW) as gentle heat — far below the 0.25 W rating of a common resistor. The LED converts $P = 2\text{ V} \times 0.0136\text{ A} \approx 27\text{ mW}$ partly into light. Always check that real power stays below each component's rating.

2.6 Tracing the current through your board

Follow conventional current around your actual design: (1) It enters at screw terminal J1, pin 1, at 5 V. (2) It travels along a copper trace to R1, where friction drops the pressure from 5 V to 2 V while limiting the flow to 13.6 mA. (3) It enters the LED's anode (the + leg, the longer wire), crosses the light-emitting junction, drops the remaining 2 V, and exits the cathode (flat side of the package). (4) Now at 0 V, it returns along a trace to J1 pin 2 (GND) and back into the power supply, completing the loop. The same 13.6 mA flows through every point of this path.

Note the LED's direction matters: it is a diode, a one-way valve. Installed backwards it simply blocks the current and nothing lights — one of the most common beginner assembly mistakes, and harmless at this voltage.

2.7 Drops happen all at once — order does not matter

A tempting but wrong mental picture: "5 V arrives at the first component, it eats some, and passes the remainder along to the next." The proof it is wrong: swap R1 and the LED so the LED comes first in the loop — the circuit behaves identically. Same current, same drops, same brightness. In a series loop there is no "first": when the loop closes, all voltage drops establish themselves simultaneously, in nanoseconds, around the entire loop at once.

So how are the drops decided? The two components play different roles. The **LED is rigid**: its datasheet forward voltage (~2 V for red) is essentially fixed whenever it conducts, whether the current is 5 mA or 20 mA. The **resistor is flexible**: its drop is $V = I \times R$, so it depends on the current. The resistor therefore absorbs whatever voltage is left over, and the loop current automatically settles at exactly the value that balances the books: $\text{leftover} = 3\text{ V}$, so $I = 3 \div 220 = 13.6\text{ mA}$. The resistor is the shock absorber of the loop.

Key idea: Kirchhoff's Voltage Law is not a design goal you must arrange — it is a law of physics that is ALWAYS satisfied, in every circuit, automatically. The real design question is: what current makes it true, and can every component survive that current?

2.8 Can the circuit live without the resistor?

Remove R1 and Kirchhoff's law still holds — by forcing the LED to drop the entire 5 V. That is where disaster strikes. Unlike a resistor, an LED's appetite for current is roughly *exponential* above its forward voltage: at 2.0 V it sips ~15 mA, at 2.5 V it may gulp hundreds of mA, and forced to 5 V it behaves almost like a plain wire — drawing the maximum the supply can deliver, converting it to heat ($P = V \times I$ becomes huge), and dying in a fraction of a second. The circuit does not fail to work without the resistor; it works lethally well for about a millisecond. The danger was never "leftover voltage with nowhere to go" — it is the runaway current the loop settles at.

A resistor is not always required — but **current must always be limited by something**. That something can be: a series resistor (our case); a supply that itself delivers only the right voltage and current (a dedicated LED driver); or the component's own internal regulation — a microcontroller chip, for example, manages its internal currents and connects directly to 5 V with no series resistor.

One related instinct to refine: real boards whose components need different voltages do *not* chain them behind resistors, feeding the highest-voltage part first and dropping the rest. A resistor's drop changes whenever the current changes ($V = I \times R$), which makes it a terrible voltage supplier for anything with varying demand. Instead, boards use **voltage regulators** to create separate stable rails (5 V, 3.3 V, 1.8 V...), and each component connects to the rail it needs. That is a later chapter — for now, remember: drops sum to the supply automatically; your job is choosing parts so the resulting current is safe.

Chapter 3 — Reading the schematic and your first PCB

3.1 What the schematic symbols mean

Your KiCad schematic contains: **J1**, a 2-position screw terminal where the external 5 V supply connects; **R1**, the 220 Ω resistor (a rectangle or zigzag symbol); **D1**, the LED (a triangle pointing at a bar — the triangle shows the allowed current direction, the arrows indicate emitted light); and the **+5V** and **GND** power symbols, which are not components at all but named nets — labels that connect every point carrying the same name.

3.2 Nets, rails, and PWR_FLAG

A *net* is one electrically connected set of points. Power symbols like +5V and GND let you avoid drawing long wires everywhere: every +5V symbol on the sheet is the same net. The two PWR_FLAG symbols in your schematic tell KiCad's electrical rules checker (ERC) that these nets really are driven by a power source — without them, ERC warns that the power pins are not driven, because it cannot know a real supply will be screwed into J1.

3.3 From schematic to physical board

The workflow you completed is the same one used for every professional board: draw the schematic (the logical design) → assign footprints (the physical land patterns each part solders onto) → route the PCB (draw copper traces connecting the pads) → run the design rule check (DRC) → export Gerber files (B_Cu/F_Cu copper layers, Mask, Silkscreen, Edge_Cuts) and drill files (PTH/NPTH), which are the manufacturing instructions a PCB factory reads.

3.4 What each exported file is

File	Meaning
F_Cu / B_Cu .gbr	Copper traces on front / back of the board
F_Mask / B_Mask .gbr	Solder mask — the green coating, with openings at pads
F_Silkscreen / B_Silkscreen .gbr	White printed labels (R1, D1, J1)
Edge_Cuts .gbr	The board outline to be cut
PTH .drl	Plated through-holes (copper-lined, conduct between layers)
NPTH .drl	Non-plated holes (mounting holes, no copper)
.csv	Bill of materials / placement data

3.5 Chapter summary

Voltage is the push (a pressure difference, measured against GND = 0 V). Current is the flow of electrons and is the same at every point of a series loop. Resistance limits the flow. Ohm's Law ($V = I \times R$) and the voltage budget (Kirchhoff) let you calculate everything in your circuit: 5 V supply – 2 V LED = 3 V across R1, giving $3 \div 220 = 13.6$ mA. The resistor exists to protect the LED, and the whole thing only works because J1 closes the loop back to the supply.

3.6 Exercises (answers at your multimeter)

1. You swap R1 for $470\ \Omega$. What is the new current? (Answer: $3 \div 470 \approx 6.4\ \text{mA}$ — dimmer.)
2. You power the board from 9 V instead of 5 V with the same red LED. What resistor keeps the current at $\sim 14\ \text{mA}$? (Answer: $(9 - 2) \div 0.014 = 500\ \Omega \rightarrow$ use $510\ \Omega$, the nearest standard value.)
3. A blue LED ($V_f = 3.2\ \text{V}$) replaces the red one with $R1 = 220\ \Omega$ at 5 V. New current? (Answer: $(5 - 3.2) \div 220 \approx 8.2\ \text{mA}$ — it works, slightly dimmer.)
4. Why does nothing light if the LED is soldered backwards, and is anything damaged? (Answer: a diode blocks reverse current; at 5 V it is safely below the LED's reverse breakdown, so no damage — just flip it.)